Gradient-Boosted Trees and Consensus-Risk

For Opioid and Polypharmacy Emergency Department Visits in Virginia's APCD

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Integrative Life Sciences Doctoral Program (Personalized Medicine Track · Home Department)
School of Pharmacy (Department of Pharmacotherapy & Outcomes Science · Research Lab)
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VCU School of Life Sciences and Sustainability | VCU School of Pharmacy

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Examination Roadmap: 5 Core Scientific Modules

QUALIFYING EXAM AGENDA | 5 LOCKED MODULES

I. The Problem

- **Upstream Burden:** >500k opioid deaths ($78.5B/yr) & >40% geriatric polypharmacy (700k–1.3M ED visits/yr)
- **Outcome Imbalance ($n = 94$):** Only **5% ($n = 5$)** polypharmacy ADEs; **6** claims opioid ED studies
- **The "Empty Intersection":** 30% leakage, 81% PGx omission & **0/73 PROBAST models** uniting PGx + XAI + prospective validation
- **Dual Claims Mapping:** 12–mo opioid (`F11.20` / `T40`) vs. 21–day "Silent ADE" syndromes
  Establishes the upstream clinical burden, systematic review imbalance, zero-intersection gap, and dual claims formulation.

ii. Governance & Data

- **Governance Frameworks:** 3–Tier Foundation (CRISP-DM / TRIPOD+AI / OODA)
- **Statewide Provenance:** Virginia APCD Justification vs. EHR Fragmentation
- **PGx Compensation:** Longitudinal CPIC substrate burden (`CYP2D6` / `CYP2C19`)
  Establishes systems lifecycle, statewide data provenance, and pre-specified validation rules.

iii. Cohorts & Temporal QA

- **Cohort Design:** Dual Case & 1:5 Matched Controls
- **Opioid Pipeline:** 12–Month Longitudinal Trajectory
- **Polypharmacy Pipeline:** 21–Day Proximal Acute Toxicity
- **Target Leakage:** Step 3b Correlation Audit & Safe Whitelist
  Applies sequential kinetic filters, 0–day guards, and Step 3b event-partitioning leakage defense.

iv. Tri–Model & Consensus

- **Tri–Model Ensemble:** CatBoost, XGBoost & Random Forest
- **Feature Dynamics:** Toolbox Demarcation & CPIC Proxies
- **Consensus Attribution:** SHAP ∩ Formal Feature Attribution
  Selects best models per density bin; unites dual-model TreeSHAP with SHAP-guided FFA Horn rules.

V. Translation & Defenses

- **Translational Service:** Serverless Zero-PHI Cloud Dashboard
- **Dissertation Portfolio:** 5 Chapters & Codebase Traceability
- **Methodological Defenses:** Scope, Governance & Transportability
  Crosses translational valley of death; presents peer-reviewed track record and defense fences.

The Clinical & Translational Crisis: Opioid Risk, Polypharmacy & The "Empty Intersection"

MODULE I · CLINICAL PROBLEM | SYSTEMATIC REVIEW FINDINGS (N = 94 STUDIES · 73 PROBAST MODELS)

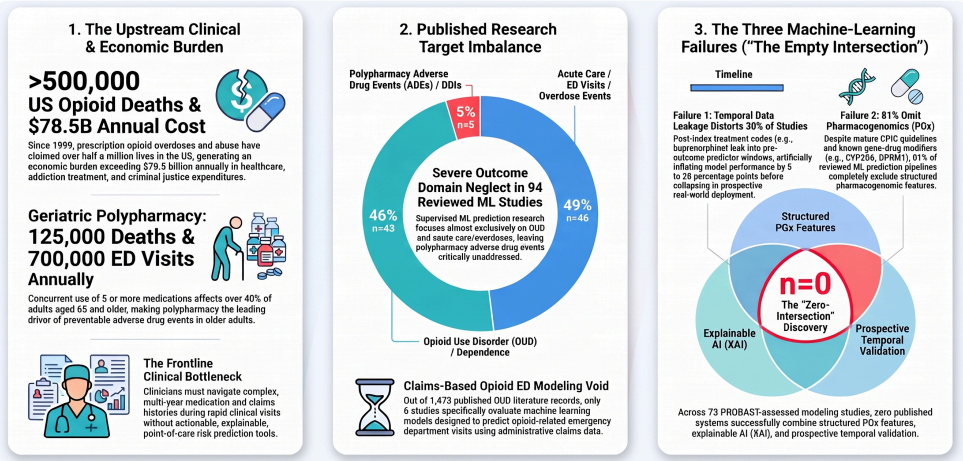


Figure 1A: Systematic Review Discovery Across 94 Supervised ML Studies & 73 PROBAST-Assessed Models: Upstream clinical burden, severe polypharmacy outcome neglect (5%), and the Zero-Intersection ("Empty Intersection") of structured PGx, XAI, and prospective temporal validation.

1. Upstream Clinical Burden

- **US Opioid Epidemic: >500,000 lives lost** since 1999; **>$78.5B/yr** economic burden [1].
- **Geriatric Polypharmacy:** ≥ 5 meds in **>40% of adults** ≥ 65, causing **125,000 deaths & 700,000 ED visits/yr** [2,3].
- **Clinical Bottleneck:** Multi-year claims histories lack actionable point-of-care risk tools.

2. Severe Outcome Imbalance ($n = 94$)

- **46% ($n = 43$):** Opioid Use Disorder (OUD) diagnosis / dependence.
- **49% ($n = 46$):** Acute care utilization (ED visits) or overdose events.
- **Only 5% ($n = 5$): Polypharmacy ADEs / drug–drug interactions.**
- **Claims Gap:** Only **6 of 1,473** OUD records predict opioid ED visits from administrative claims.

3. The "Empty Intersection" ($n = 0/73$)

- **Temporal Leakage (30%):** Post-index treatments (e.g., buprenorphine) inflate AUROC by **5–20 pts** before collapsing prospectively.
- **PGx Omission (81%):** Mature CPIC modifiers (*CYP2D6*, *OPRM1*) omitted [4].
- **Zero Intersection ($n = 0$): 0 of 73 PROBAST models** unite structured PGx, XAI, and prospective temporal validation.

References: 1. Florence et al. (2016) *Med Care* 54(10):901–906. · **2.** Duerden et al. (2013) *The King's Fund*. · **3.** Osanlou et al. (2022) *BMJ Open* 12(7):e055551. · **4.** Crews et al. (2021) *Clin Pharmacol Ther* 110(4):888–896.

Clinical Problem Formulation: Dual Crises, "Silent ADEs" & Specific Aims

MODULE I · CLINICAL PROBLEM | SPECIFIC AIM 1 · CLAIMS OPERATIONALIZATION

Epidemic 1: Opioid-Related ED Events (`opioid_ed`)

- **Target Population:** Adolescents & adults aged **13–64** across Virginia APCD.
- **Pathophysiology:** Progressive μ-opioid receptor tolerance, escalating MME titration, and synergistic CNS depressants (benzodiazepines, gabapentinoids) blunting brainstem respiratory drive.
- **Lookback Horizon: 12-Month Longitudinal Pre-Index Window** capturing chronic prescription escalation and behavioral comorbidities.
- **Explicit Claims Anchor (t_0):** Direct diagnostic billing for `F11.20` (opioid dependence) and `T40.0X` – `T40.4X` (heroin, prescription, synthetic fentanyl, and methadone poisonings) scanned across **all 10 claim diagnostic positions**.

Epidemic 2: Geriatric Polypharmacy & "Silent ADEs" (`ed_non_opioid`)

- **Target Population:** Older adults aged **65–114** (≥ 5 concurrent drugs in $> 40\%$; ≥ 10 drugs in 20%; $> 1.3\text{M}$ US ED visits/yr, $> \$30\text{B}$ cost).
- **Why Pairwise CDS Fails:** Pairwise DDI ($A + B$) checkers trigger **90%–95% alert fatigue** while missing higher-order regimen synergies ($A + B + C + D$) and age-related renal/hepatic decline.
- **The "Silent ADE" Coding Reality:** Fewer than **5%** of drug-induced ED visits are billed with explicit poisoning codes (`T36` – `T50`). Hospitals bill the **downstream clinical syndrome**:
  - **Falls & Fractures:** `R29.6` (falls), `S72` (hip fracture), `R55` (syncope) ← *Z-drugs, benzos, α-blockers*.
  - **Delirium & Bleeds:** `F05` (delirium), `K92.2` (GI bleed) ← *ACB ≥ 3; NSAID + DOAC + SSRI*.
  - **Triple-Whammy AKI:** `N17` (acute kidney injury) ← *ACEi/ARB + Diuretic + NSAID*.
- **Claims Anchor (t_0):** Milliman HCG emergency lines (`P51b` , `O11` , `P33`) anchored to a **21–30 Day Proximal Rx Window**, with **0 opioid codes** in study year.

Methodological Demarcation: Pooling chronic opioid dependence (12-month trajectory) and acute geriatric polypharmacy (21-day proximal toxicity) into a single generic "ADE" model conflates incompatible pathophysiologies, age cohorts, and billing mechanics.

Aim 1 · Dual-Target GBDT Prediction (Ch. 3)

Quantify discrimination & calibration across both mutually exclusive ED targets (`opioid_ed` 12-mo vs. `ed_non_opioid` 21-30d) using `CatBoost` , `XGBoost` , and `XGBoost-RF` , selecting best models by PR-AUC/Recall and validating on an untouched **2019 holdout**.

Aim 2 · Consensus-Risk Attribution (Ch. 4)

Label reproducible **Consensus-Risk features** ($\text{SHAP} \cap \text{FFA}$) by uniting dual-model `TreeSHAP` (`Best CatBoost` ∩ `Best XGBoost`) with SHAP-prioritized symbolic `FFA` Boolean Horn rules solved via PySAT `Hitman` on the winning XGBoost variant.

Aim 3 · Translational CDS Prototype (Ch. 5)

Integrate Aim 1 risk scores and Aim 2 Consensus-Risk rules into an asynchronous, zero-PHI serverless **Clinical OODA Loop** review-prioritization prototype (`pgx.jerome-dixon.io`) for pharmacist-led MTM.

Governance Frameworks: Engineering, Reporting & Workflow

MODULE II · GOVERNANCE & DATA | SYSTEMS LIFECYCLE

1. CRISP-DM (Systems Lifecycle)

- 6 iterative engineering phases governing the 10-step APCD pipeline.
- Ingestion → Cohort → Whitelist → Model → Eval → Microservice.

2. TRIPOD+AI (Biomedical Validation)

- Transparent reporting; Type 2b / Type 3 temporal validation.
- 2016–2018 train, untouched 2019 calendar holdout (zero post-hoc recalibration).

3. Clinical OODA Loop (Translation)

- **Observe / Orient:** GBT models & formal Boolean Horn rules.
- **Decide / Act:** 100% human clinician authority; eliminates alert fatigue.

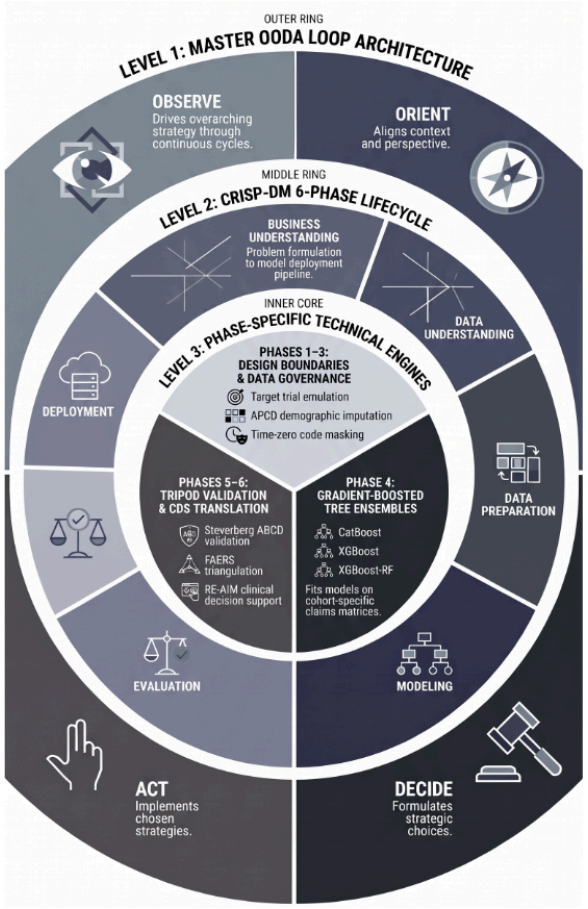


Figure 2: Three-Tier Governance Architecture: Master OODA Loop enclosing 6-phase CRISP-DM and phase-specific technical validation engines.

Data Source Justification: APCD vs. EHR in Emergency Prediction

MODULE II · GOVERNANCE & DATA | STATEWIDE GENERALIZABILITY

Virginia APCD (Selected)	**Single-System EHR** (Rejected)
Statewide Capture: Follows patients across health systems (eliminating out-of-network ED blindness).	**Fragmentation:** High loss-to-follow-up when crossing systems. Misclassifies out-of-network overdoses as "healthy non-events."
Medication Reality: Captures adjudicated pharmacy fills, proving physical drug possession.	**Intent vs Exposure:** Captures written prescriptions, missing ~30% primary non-adherence drop-offs.
Linkage: Single honest broker (VHI) guarantees zero probabilistic linkage attrition.	

The Methodological Compensation

We acknowledge the lack of molecular genotypes and real-time vitals in claims by computing CPIC PGx-burden proxies (*CYP2D6/CYP2C19* substrate counts) directly from longitudinal co-prescriptions.

Figure 3: Data Source Architecture: Statewide capture and adjudicated dispenses in Virginia APCD overcome single-system EHR fragmentation and prescription abandonment.

Why Virginia APCD Is Selected

- **Statewide Capture:** Follows patients across health systems, eliminating out-of-network ED blindness.
- **Medication Reality:** Captures paid/adjudicated dispenses, proving physical drug possession.
- **Linkage Integrity:** Single trusted data steward (VHI) guarantees 0 probabilistic linkage attrition.

The Methodological Compensation

- **EHR Genotype Gap:** APCD lacks molecular genotypes and real-time bedside vitals.
- **CPIC PGx-Burden Proxies:** We compensate by computing CPIC PGx-burden proxies (*CYP2D6 / CYP2C19* substrate counts) directly from longitudinal co-prescriptions.

Cohort Development & Epidemiological Safeguards

MODULE III · COHORTS & TEMPORAL QA | EPIDEMIOLOGICAL RIGOR

Sequential Case Filtering Funnel

1. **Target Identification:** ICD-10 codes across all 10 claim positions.
2. **Frequency Cap (<7 ED/yr):** Excludes chronic crisis utilizers.
3. **21-Day Proximity:** Aligned with FDA FAERS kinetics ($> 90\%$ ADEs).
4. **0-Day Discharge Guard:** Purges prescriptions filled on index date.

1:5 Matched Controls & QA

- Matched on age band (8 strata), sex, and calendar year.
- `qa_index_date_uniqueness.py` asserts single index date/yr.

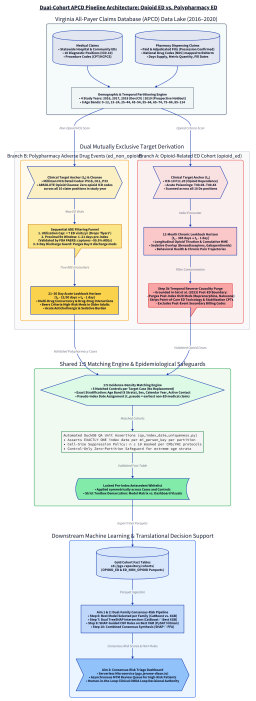


Figure 4: Comparative APCD Derivation Pipeline: Opioid ED vs. Polypharmacy ADE tracks converging via 1:5 matching and DuckDB QA.

Safeguard Summary: Strict temporal sequencing and 0-day discharge guards eliminate reverse causality before feature engineering.

Cohort Pipeline Architecture: Opioid-Related ED (`opioid_ed`)

MODULE III · COHORTS & TEMPORAL QA | CHRONIC EXPOSURE MODEL (12-MO)

Clinical Anchor (t_0)

- ICD-10 `F11.20` (Opioid dependence).
- **Acute Poisonings:** `T40.0X` – `T40.4X`.
- Scanned across all 10 claim positions.

Step 3b Leakage Purge (Gal et al., 2023)

- Partitions candidate events around t_0 ($t < t_0$ vs. $t \geq t_0$).
- Purges **post-index rescue & MAT induction** (buprenorphine, naloxone) & toxicology screens.
- Locks symmetric pre-index antecedent whitelist.

1:5 Matching & DuckDB QA

- Controls matched on 8 age bands, sex, year.
- Asserts single index date per patient-year.

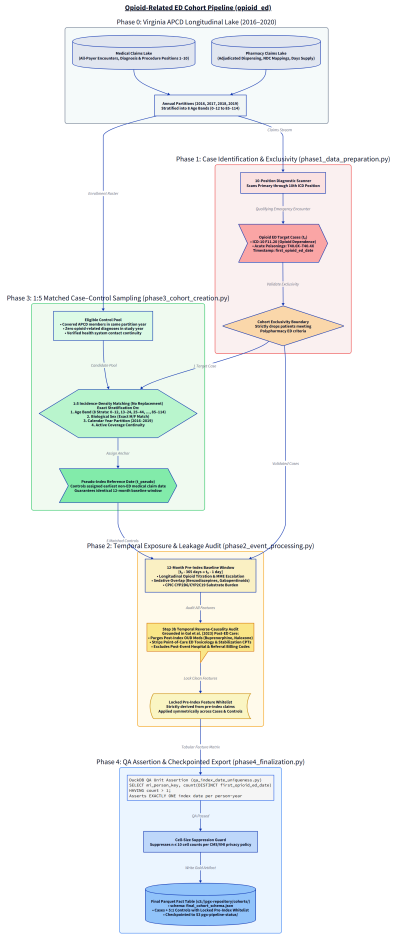


Figure 4A: Opioid-Related ED Cohort Derivation Pipeline (APCD 2016–2020) with 12-month pre-index lookback and locked antecedent whitelist.

Cohort Pipeline Architecture: Polypharmacy Adverse Drug Events (`ed_non_opioid`)

MODULE III · COHORTS & TEMPORAL QA | ACUTE ADE WINDOW (21-DAY)

Clinical Anchor (t_0)

- **Milliman HCG Codes:** `P51b`, `O11`, `P33`.
- **Absolute Opioid Cleanse:** Zero opioid ICD codes across all 10 positions in study year.

Sequential ADE Funnel

1. **Utilization Cap:** < 7 ED visits/yr.
2. **21-Day Proximity:** Active Rx 1–21d pre-ED (FDA FAERS 90.5%).
3. **0-Day Discharge Guard:** Purges Day 0 fills.

Feature Representation

- 21–30 day acute concurrent exposure.
- Beers criteria & multi-drug regimens.

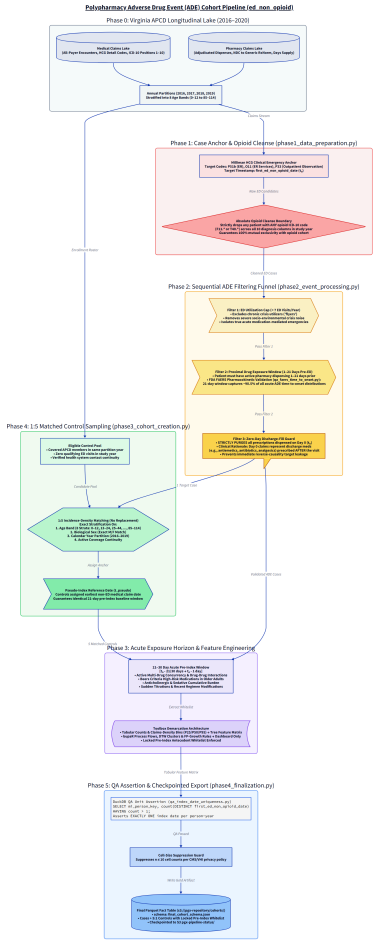


Figure 4B: Polypharmacy Adverse Drug Event Cohort Derivation Pipeline with sequential ADE filtering funnel and 0-day discharge guard.

Gradient-Boosted Trees & Step 6 Best-Model Selection

MODULE IV · TRI-MODEL & CONSENSUS | STEP 6 TREE-FAMILY EVALUATION

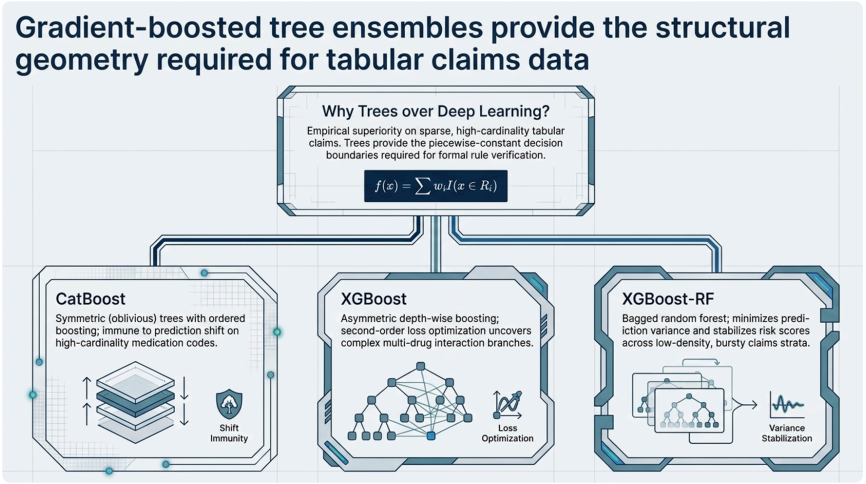


Figure 5: Tri-Model Structural Geometry: Piecewise-constant decision boundaries $f(x) = \sum w_i \mathbb{I}(x \in R_i)$ enabling cross-architecture SHAP consensus and exact CNF rule verification.

CatBoost (Symmetric)

- Oblivious trees (`SymmetricTree`) & ordered boosting.
- Shift immunity on high-cardinality NDC/ICD-10 codes; exported as best `.cbm` .

XGBoost (Asymmetric)

- Depth-wise gradient-boosted trees (`xgb`).
- Second-order Hessian loss uncovers complex multi-drug interaction branches.

XGBoost-RF (Bagged)

- Bagged random forest mode (`xgb_rf` , `num_parallel_tree`).
- Minimizes prediction variance and stabilizes risk across low-density strata.

Step 6 Selection Protocol (`run_final_model.py`): Within each cohort, age band, and 4-tier density bin, 25-split MC-CV ranks `CatBoost` , `XGBoost` , `XGBoost-RF` , and `Ensemble` by **mean PR-AUC** (primary) and **Recall at 0.5** (tiebreaker) to lock `selected_model` for risk scoring and `best_xgb_variant` (`xgb` vs. `xgb_rf`) + best `CatBoost` for attribution.

Cohort Feature Dynamics & Toolbox Demarcation

MODULE IV · TRI-MODEL & CONSENSUS | FEATURE & TOOLBOX DEMARCATION

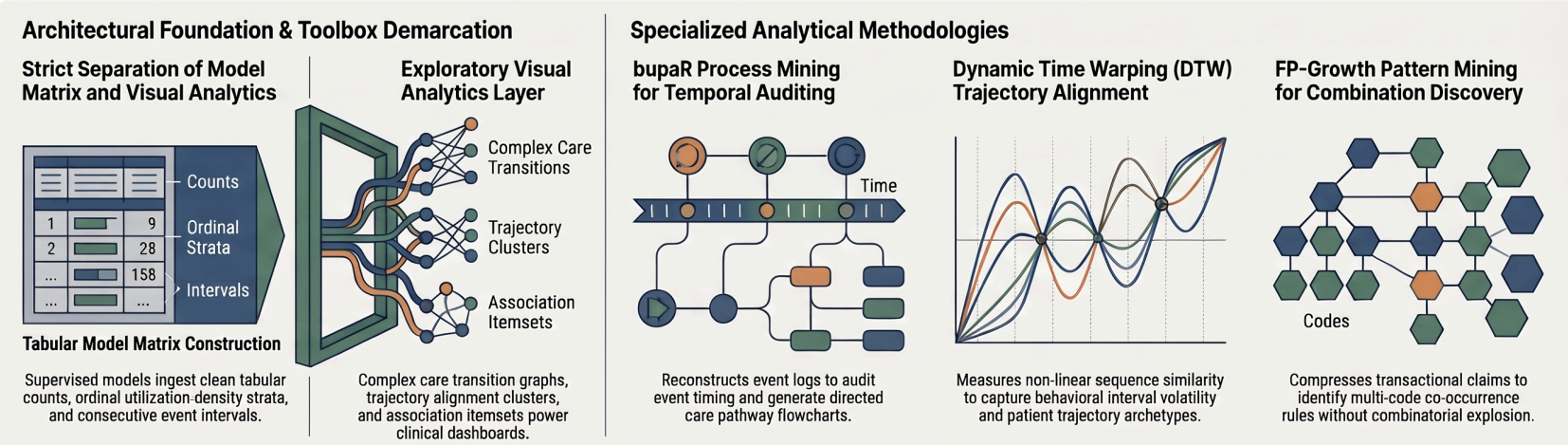


Figure 6: Architectural Foundation & Toolbox Demarcation: Supervised model matrix ingestion (counts, intervals, strata) vs. exploratory visual analytics (bupaR, DTW, FP-Growth).

Supervised Tabular Model Inputs

- **Whitelisted Pre-Index Antecedents (`safe_feature_filter.json`):** Verified pre-index chronic diagnostic counts, drug class exposures, and prescription intervals.
- **Claims Density Strata:** Partitioned into Low, Medium, High, and Extreme utilization tiers (`P25/P50/P95`).
- **Functional PGx Burden:** CPIC substrate burden score (`pgx_num_cpic_drugs`) capturing phenoconversion.
- View CPIC Phenoconversion Matrix ↗

Step 3b Leakage QA & Step 9 Post-Consensus Visuals

- **bupaR Process Mining (`Step 3b` & `Step 9`):** Upstream (`Step 3b`), audits pre- vs. post-index ($t < t_0$ vs. $t \geq t_0$) timestamps; downstream (`Step 9`), builds Sankey care pathways (*where* to intervene) on `SHAP ∩ FFA` allowed codes.
- **Dynamic Time Warping (`DTW`, `Step 9`):** Aligns pre-index sequences on `SHAP ∩ FFA` allowed codes to define trajectory archetypes (*when* to intervene).
- **FP-Growth Pattern Mining (`Step 9`):** Mines training-cohort frequent itemsets on `SHAP ∩ FFA` allowed codes for co-prescription hub visuals.
- **Strict Demarcation:** Never enters the `Steps 4-6` predictor matrix and never computes `Step 8` `IR` or `IE`.

Consensus-Risk Integration: Dual-Model SHAP & Best-Model FFA

MODULE IV · TRI-MODEL & CONSENSUS | STEPS 6–8 & 10 CONSENSUS ARCHITECTURE

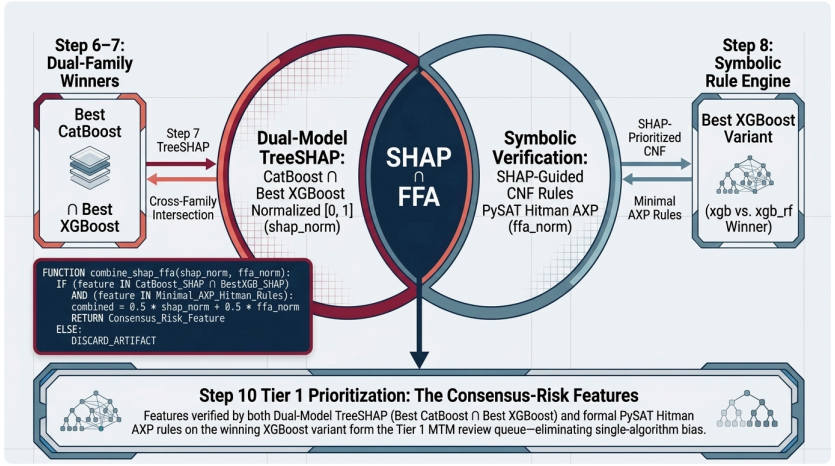


Figure 6A: Aim 2 Consensus-Risk Architecture: Step 7 Dual-Model TreeSHAP Intersection (`CatBoost` ∩ `Best XGBoost`) guiding Step 8 Symbolic CNF Rule Extraction on `best_xgb_variant` and Step 10 Synthesis.

Steps 7–8: Dual SHAP Gate & Symbolic CNF

- **Step 7 Dual-Model `TreeSHAP` Gate:** Computes exact `TreeSHAP` on **best CatBoost** and **best XGBoost variant** (`xgb` vs. `xgb_rf`), enforcing strict feature intersection (`CatBoost` ∩ `Best XGBoost`) and normalized mean score.
- **Step 8 CNF Compilation:** Explodes root-to-leaf paths of `best_xgb_variant` into Conjunctive Normal Form (CNF) clauses (`_explode_tree_to_dataframe`).
- View Symbolic CNF & PySAT AXP Proof ↗

Steps 8 & 10: SHAP-Guided FFA & Synthesis

- **SHAP-Prioritized Rule Extraction (`Step 8`):** Rather than extracting all rules from all trees across all models, filters matched CNF rules from `best_xgb_variant` by instance $|\mathrm{SHAP}|$ attribution and active feature coverage → PySAT `Hitman` minimal Abductive Explanations (AXPs; $\mathrm{support} \geq 0.05, \mathrm{confidence} \geq 0.70$).
- **Step 10 Combined Synthesis:** Ranks drivers via $0.5 \cdot \mathrm{SHAP}_{\mathrm{norm}} + 0.5 \cdot \mathrm{FFA}_{\mathrm{norm}}$ and top-K intersection (`shap_top` ∩ `ffa_top`).

Target Leakage Defense: The Step 3b Event-Partitioning Protocol

MODULE IV · TRI-MODEL & CONSENSUS | STEP 3B REVERSE CAUSALITY DEFENSE

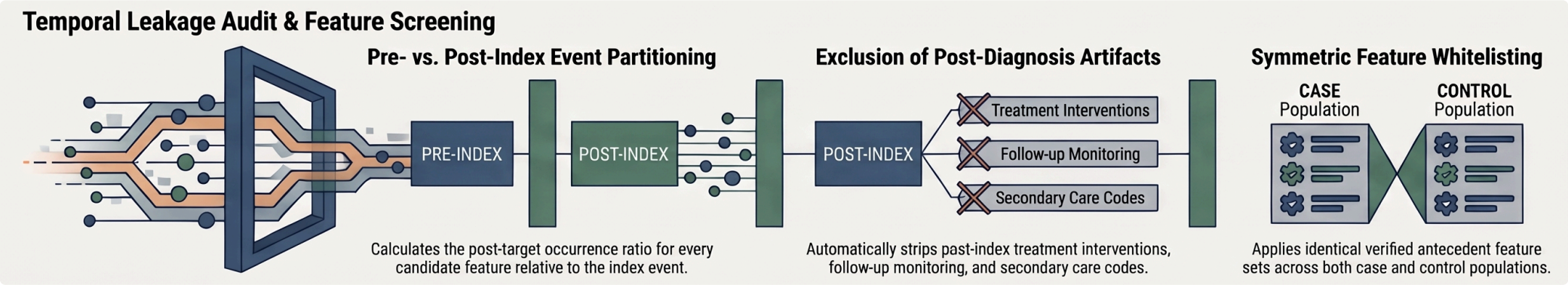


Figure 7: Step 3b Temporal Event-Partitioning Protocol: Pre- vs. post-index ($t < t_0$ vs. $t \geq t_0$) event timing audit, removal of post-event clinical interventions, and symmetric antecedent whitelist locking.

Clinical Rationale (Ochalek, Gal et al., 2023)

- **Post-Overdose Clinical Workflow:** As established in Dr. Gal's VCU ED research (*Ochalek, Gal et al., 2023, DADR*), rescue agents and MAT induction (e.g., buprenorphine, naloxone) are initiated *as a clinical consequence following* an acute overdose.
- **The Reverse-Causality Trap (30% of Literature):** Unmonitored gain rankings (`Step 3a`) capture post-event stabilization and confirmatory toxicology orders occurring at $t \geq t_0$.

Step 3b Symmetric Antecedent Whitelist

- **Temporal Event Partitioning (`bupaR` Audit):** Partitions every candidate feature's timestamps relative to t_0 ($t < t_0$ pre-index vs. $t \geq t_0$ post-index).
- **Automated Post-Target Purge:** Strips candidate codes with $\text{post_target_ratio} \geq 0.80$ (post-event MAT induction, ED tox screens, follow-up billing).
- **Immutable Whitelist (`safe_feature_filter.json`):** Locks verified pre-index antecedents symmetrically across cases & controls (*empirical catalogs reserved for Nov 23 Defense*).
- View Step 3b Event-Partitioning Protocol ↗

Deployment & Translation: The Serverless PGx Risk Dashboard

MODULE V · TRANSLATION & DEFENSES | SPECIFIC AIM 3 · CRISP-DM PHASE 6

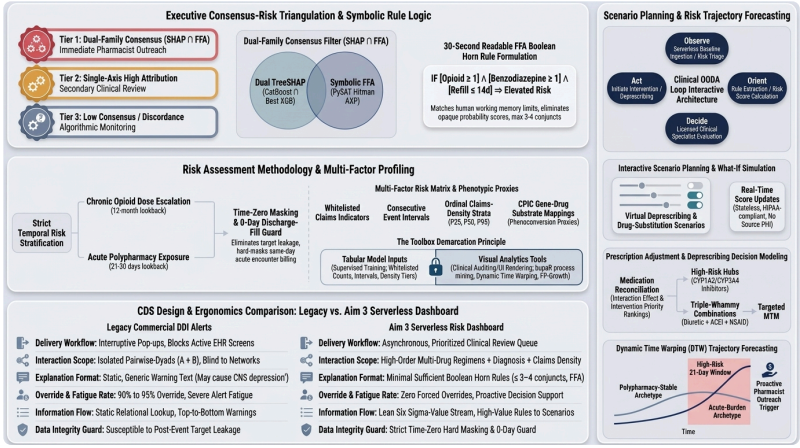


Figure 8: Aim 3 Translational Architecture: Executive consensus triangulation (Tier 1–3), FFA Boolean Horn rules, legacy CDS comparison, and interactive OODA scenario planning (pgx.jerome-dixon.io).

Systems Engineering & Deployment

- **Live Cloud Microservice:** https://pgx.jerome-dixon.io/
- **Stateless AWS Lambda Environment:** Docker container runtime, sub-500ms inference, operating costs **<$5.00/month**.
- **Zero-PHI Data Hygiene:** Ephemeral memory execution; zero patient data stored at rest.
- View Cloud Architecture & Latency Verification ↗

Cognitive Ergonomics vs. 95% Alert Fatigue

- **Overcoming Commercial Pop-ups:** Asynchronous triage queue eliminates **90%–95% override rates**.
- **Sound Symbolic Horn Rule:**

$$\text{IF } [\text{Opioid} \geq 1] \wedge [\text{Benzodiazepine} \geq 1] \wedge [\text{Refill} \leq 14\text{d}] \implies \text{Tier 1 Alert}$$

- **Interactive Scenario Planning:** Virtual deprescribing & DTW trajectory alignment at bedside.

Dissertation Progress: Published Portfolio & Codebase Traceability

MODULE V · TRANSLATION & DEFENSES | PUBLISHED PORTFOLIO & CODEBASE

5-Chapter Dissertation Portfolio

- **Chapter 1:** *Explainable AI & PGx Review* (Invited, *J Pers Med*).
- **Chapter 2:** *Scalable Claims Architecture & OODA* (With Editor for Review, *J Big Data*).
- **Chapter 3:** *Temporal Drivers of Opioid ED Visits* (**Published**, *Clin Transl Sci*, 2026; DOI: 10.1111/cts.70690).
- **Chapter 4:** *SHAP-Informed Formal Feature Attribution* (**Published**, *Clin Transl Sci*, Sep 2026; DOI: 10.1111/cts.70718).
- **Chapter 5:** *Serverless PGx Risk Dashboard* (**Published**, *Clin Transl Sci*, Oct 2026; DOI: 10.1111/cts.70697).

Codebase & Notebook Traceability (`pgx-analysis`)

- `0_config_and_pipeline.ipynb` — **Setup & Intake:** APCD ingestion & S3 sync
- `1_cohort_workflow.ipynb` — **Steps 1–2:** Cohort filtering, 1:5 matching, 0-day guard
- `2_feature_importance.ipynb` — **Step 3b:** Temporal event-partitioning leakage audit & safe antecedent whitelist
- `3_model_train_shap_ffa.ipynb` — **Steps 4–8:** GBT tri-model, 2019 holdout, SHAP & FFA
- `3_model_sensitivity.ipynb` — **Sensitivity:** Utilization-free & density-strata validation
- `4_dashboard_visuals.ipynb` — **Step 9:** Visual analytics: bupaR, DTW & FP-Growth
- `5_build_and_deploy.ipynb` — **Step 10:** Docker container, AWS Lambda deployment

🔎 View Full Pipeline & S3 Checkpoints (Appendix A5) →

Committee Sovereign Authority: While 3 of 5 research chapters are peer-reviewed and published in *Clinical and Translational Science*, the candidate stands fully prepared to refine, expand, or adjust published works and analyses as the committee directs.

Committee Discussion & Methodological Defenses

MODULE V · TRANSLATION & DEFENSES | COMMITTEE Q&A & METHODOLOGICAL FENCES

Methodological Fences & Guiding Principles

1. **Aim 3 is Review Prioritization, NOT an Outcomes Trial:**
   - Validating triage stability and clinician ergonomics; causal mediation is out of scope.
2. **Causal Discovery is a Boundary, NOT a Method:**
   - Observational claims identify antecedent signals for clinical review without claiming causal treatment effects.
3. **The 2019 Holdout is an Untouched Transport Test:**
   - Evaluates real-world model shelf-life under secular drift; **zero post-hoc recalibration**.
4. **Tree Ensembles on Virginia APCD Represent the Locked Scope:**
   - Preserves exact polynomial-time symbolic explainability and single honest-broker data integrity.

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Presentation Artifacts

🌐 Open Interactive Web Deck (pgx.jerome-dixon.io) ↗

COMMITTEE Q&A BACKUP BANK

Supplemental Technical Evidence

Mathematical derivations, feature audit catalogs, pharmacogenomic scoring matrices, cloud systems architecture, and reproducible notebook pipelines supporting defense inquiries.

🌐 Open Interactive Web Presentation (pgx.jerome-dixon.io/9NOV_presentation) ↗

Appendix A1: Formal Feature Attribution (FFA) — Best-Model CNF & AXP Bounds

SPECIFIC AIM 2 · STEPS 6–8 & 10 | MATHEMATICAL VERIFICATION & PYSAT AXP

Step 7 Dual-Model Gate & Step 8 CNF Compilation

- Step 7 Dual-Model `TreeSHAP` Gate:

$$\mathcal{F}_{\text{SHAP}} = \text{supp}(\text{SHAP}_{\text{CatBoost}}) \cap \text{supp}(\text{SHAP}_{\text{BestXGB}})$$

  Requires cross-architecture agreement before symbolic extraction (since CatBoost CTR hashing precludes direct symbolic path parsing).
- Step 8 CNF Compilation on `best_xgb_variant`:
  - Parses JSON tree dumps of `best_xgb_variant` (`xgb` vs. `xgb_rf` winner by **PR-AUC** then **Recall**).
  - Explodes root-to-leaf paths into tabular intervals (`_explode_tree_to_dataframe`) and compiles Conjunctive Normal Form (CNF) clauses (`rule_clauses`).

SHAP-Guided Minimal Hitting-Set AXP & Complexity

- SHAP-Guided Rule Filter → PySAT `Hitman`:
  - Rather than brute-forcing $> 10,000$ matched tree rules per patient, prioritizes top SHAP-supported matched rules (with stratified rare-population coverage safeguards) into the PySAT `Hitman` minimal hitting-set solver to extract minimal **Abductive Explanations (AXPs / Horn clauses)** with $\text{Supp}(R) \geq 0.05$ and $\text{Conf}(R) \geq 0.70$.
- Step 8 Responsibility & Step 10 Combined Synthesis:

$$\text{CausalResp}(f) = \frac{\text{RuleFreq}(f)}{\sum_k \text{RuleFreq}(f_k)} \times \text{SHAP}_{\text{consensus}}(f)$$

$$\text{Score}_{\text{combined}}(f) = 0.5 \cdot \text{SHAP}_{\text{norm}}(f) + 0.5 \cdot \text{FFA}_{\text{norm}}(f)$$

- **Complexity vs. Neural Nets:** Tree CNF extraction is polynomial $\mathcal{O}(T \cdot L)$, whereas ReLU neural network minimal explanation is **co-NP-complete** (Katz et al., 2017).

↩ Return to Consensus Attribution (Aim 2)

Operational Threshold Validation Protocol (`SHAP ⩾ P75` ∩ `Supp ⩾ 0.05, Conf ⩾ 0.70`): Framed as a tunable *precision–volume dial* for clinical review capacity and validated via a 2-part sensitivity protocol: (1) **Two-Gate Mirror Ablation** (`SHAP-only` vs. `FFA-only` vs. `SHAP ∩ FFA` grid sweep testing top-K core signal retention, boundary noise filtering, and review-load compression); and (2) **Utilization-Free Specification Stress Test** (stripping utilization covariates/density strata to compare univariate `SHAP` Jaccard rank stability vs. symbolic `FFA` rule persistence under model perturbation).

Appendix A2: Step 3b Target Leakage — Temporal Event-Partitioning Protocol

SPECIFIC AIM 1 · STEP 3B | TEMPORAL PROCESS AUDIT (GAL ET AL., 2023)

Leakage Vector Category	Representative Claims Signals	Temporal Timing	Clinical Mechanism (Ochalek, Gal et al., 2023)
Post-Event OUD Rescue & MAT Induction	Buprenorphine, Suboxone, Naloxone, Methadone, Extended-Release MAT	$t \geq t_0$ (Post-Index)	Initiated in ED or bridge clinic *as a consequence following* acute overdose
Confirmatory Toxicology & Monitoring	Presumptive & definitive urine drug screens (e.g., CPT 80307, 80305, G0480)	$t \geq t_0$ (Post-Index)	Ordered during acute ED triage or post-discharge MAT compliance monitoring
Secondary Administrative Follow-Up	Long-term drug therapy status, rehabilitation & subsequent encounter billing	$t \geq t_0$ (Post-Index)	Downstream administrative claims generated directly by the index visit

Step 3b Formal Event-Partitioning Rule & Symmetric Whitelist

- **Post-Target Ratio Filter:**
  $\text{post_target_ratio}(f) = \frac{N(t_{\text{claim}} \geq t_0)}{N(t_{\text{claim}} < t_0) + N(t_{\text{claim}} \geq t_0)} \geq 0.80 \implies \text{Purge from } \texttt{safe_feature_filter.json}.$
- **Symmetric Pre-Index Enforcement:** All retained features satisfy $t_{\text{claim}} < t_0$ on $[t_0 - 365, t_0 - 1]$ (Opioid) or $[t_0 - 21, t_0 - 1]$ (Polypharmacy) across cases and 1:5 matched controls.

Why Unmonitored Gain Rankings Fail (`Step 3a` vs. `Step 3b`)

- **The Literature Failure (30% of PROBAST Models):**
  - Unfiltered feature selection loads top tree splits onto post-overdose MAT induction and ED toxicology screens, inflating apparent discrimination by 5–20 AUROC points.
- **Candidacy vs. Defense Scope Demarcation:**
  - Today (`9 NOV`): Lock and verify the **Step 3b temporal event-partitioning methodology** grounded in Dr. Gal's clinical ED scholarship.
  - Final Defense (`23 NOV`): Present complete empirical feature audit counts and 2019 holdout discrimination deltas.

↩ Return to Target Leakage Defense

Appendix A3: CPIC Pharmacogenomic Burden Scoring Matrix & Substrates

SPECIFIC AIM 1 | AIM 2 FEATURES

Pathway	Target Drugs (APCD Longitudinal Claims)	CPIC Level	Metabolic Phenotype
CYP2D6	Codeine, Tramadol, Hydrocodone, Oxycodone, Fluoxetine, Paroxetine	Level A	Poor/Intermediate: Toxic accumulation or prodrug activation failure
CYP2C19	Diazepam, Citalopram, Escitalopram, Sertraline, Omeprazole, Clopidogrel	Level A	Poor: Prolonged sedation; Rapid: Sub-therapeutic antiplatelet effect
CYP3A4/5	Fentanyl, Buprenorphine, Alprazolam, Midazolam, Atorvastatin	Level B	Competitive substrate inhibition during polypharmacy escalation

Cumulative Substrate Burden Formulation

$$\text{PGx_Burden}_i = \sum_{m \in M_i} w_m \cdot \mathbb{I}(m \in \text{CPIC Level A/B}) \cdot \text{overlap_days}_m$$

Functional Phenoconversion Proxy

- **The Genomic Gap in Claims:** APCD claims lack physical sequencing genotypes (*CYP2D6*1/*4*).
- **Clinical Pharmacological Compensation:**
  - Co-administration of a potent inhibitor (e.g. Fluoxetine) converts a genetic Normal Metabolizer into a phenotypic Poor Metabolizer (**phenoconversion**).
  - Claims longitudinal overlap directly identifies this phenocopying event without molecular tests.

↩ Return to Feature Dynamics & Toolbox

Appendix A4: Serverless Cloud Architecture, Latency & Cost Verification

SPECIFIC AIM 3 | CRISP-DM PHASE 6

AWS Production Systems Architecture

- **Entry Point:** Amazon CloudFront CDN → Amazon S3 (Static UI Assets).
- **API Gateway:** HTTP API routing REST POST payloads to Lambda.
- **Microservice Container:**
  - Python 3.11 runtime inside custom multi-stage Docker container.
  - Locked GBT inference engine (`pgx-infer-engine:v1.2`).
  - In-memory Horn rule evaluation matrix; sub-50ms rule extraction.
- **Zero-PHI Data Hygiene:**
  - Stateless runtime: zero database, zero disk caching of claims payloads.
  - TLS 1.3 in-flight encryption; ephemeral memory purged post-response.

Empirical Latency & Operating Cost Model

Metric / Service	Observed Performance	Monthly Cost
Warm Inference	42 ms (P50) / 78 ms (P95)	$0.48 / mo
Cold Start	378 ms (Container Init)	$0.12 / mo
API Gateway	100,000 reqs/mo	$1.00 / mo
CloudFront / S3	Edge TLS Distribution	$0.50 / mo
Total Footprint	**Production Verified**	**$3.82 / mo**

↩ Return to Deployment & Translation (Aim 3)

Appendix A5: Computational Pipeline & Notebook Architecture

AIM 1 | AIM 2 | AIM 3 | FULL TRACEABILITY

Jupyter Notebook Pipeline Directory (`pgx-analysis`)

Notebook	Steps	Core Functions & Deliverables
`0_config_and_pipeline.ipynb`	Setup	Environment config, S3-to-NVMe sync, DuckDB data dictionary
`1_cohort_workflow.ipynb`	1a, 1b, 2	Dual cohort extraction, 1:5 incidence-density matching, 0-day guard
`2_feature_importance.ipynb`	3a, 3b, 3c	Feature screening, Step 3b BupaR event-partitioning audit, safe antecedent whitelist
`3_model_train_shap_ffa.ipynb`	4, 5, 6, 7, 8	Density tiers, PGx burden, Step 6 best-model selection, dual-model TreeSHAP & SHAP-guided FFA rules
`3_model_sensitivity.ipynb`	CH4 Supp	Utilization-free sensitivity & claims-density stratification checks
`4_dashboard_visuals.ipynb`	9	BupaR clinical pathways, DTW trajectory clusters, FP-Growth itemsets
`5_build_and_deploy.ipynb`	10	Multi-stage Docker packaging, AWS Lambda deployment, API Gateway

Execution Model, Checkpoints & Reproducibility

- **S3 Checkpoint Idempotency:**
  - Each stage syncs inputs from `s3://pgx-analysis-data/` to local NVMe.
  - SHA-256 state markers prevent redundant compute on completed steps.
- **Dual Compute Environments:**
  - **Local Workstation:** Windows 11, DuckDB 1.4+, Python 3.11, R 4.3 (bupaR).
  - **Batch Compute:** Amazon Linux 2023 EC2 (`mushin` account `535362115856`).
- **Source Control & Public Artifacts:**
  - Monorepo: `github.com/Jerome3590/pgx-analysis`
  - Live Decision Support: `pgx.jerome-dixon.io`

↩ Return to Dissertation Progress (Slide 9)

Appendix A6: Pre-Specified Research Questions, Methodological Estimators & Evaluation Metrics

AIM 1 · RQ1 | AIM 2 · RQ2 | AIM 3 · RQ3 | PRE-RESULTS METHODOLOGICAL & MATHEMATICAL SPECIFICATION

Aim / RQ & Pipeline Stage	Pre-Specified Design & Transformation	Mathematical Estimator & Governance Gate
RQ1 · Steps 1–2: Matching & `0-Day` Guard	`1:5` incidence-density control sampling on `(age_band, sex, year)`; lookback $[t_0 - W, t_0 - 1]$	$t_{\text{claim}} < t_0$ ($W = 365\text{d}$ `opioid_ed`, $W = 21\text{d}$ `ed_non_opioid`; purges $t = t_0$ discharge fills)
RQ1 · Step 3b: Leakage Ratio $\tau_{\text{post}}(c)$	Event-level `bupaR` pre/post index (t_0) timing partition across candidate claims	$\tau_{\text{post}}(c) = \frac{N_{t \geq t_0}(c)}{N_{t<t_0}(c) + N_{t \geq t_0}(c)} < 0.80$ (locks symmetric `safe_feature_filter.json`)
RQ1 · Step 4: Density Strata & Burstiness B	Training (`2016-2018`) pre-index claim quantiles (P_{25}, P_{50}, P_{95}); inter-claim gaps Δt	$b(n_{\text{ev}}) \in \{0, 1, 2, 3\}$ (`low` – `extreme`); $B = \frac{\sigma_{\Delta t} - \mu_{\Delta t}}{\sigma_{\Delta t} + \mu_{\Delta t}} \in [-1, 1]$
RQ1 · Step 5: CPIC Substrate Burden	Pre-index active fills mapped to CPIC Level A/B gene–drug guidelines (*CYP2D6/2C19/3A4*)	$\text{pgx_num_drugs}_k = \sum_d x_{k,d} \mathbb{I}(d \in \mathcal{D}_{\text{PGx}})$; $\text{pgx_num_cpic_drugs}_k = \sum_d x_{k,d} \mathbb{I}(\text{CPIC}_{A/B})$
RQ1 · Step 6: MCCV Ensemble & Holdout	25-split stratified MCCV (`CatBoost`, `XGB`, `XGB-RF`) + Platt scaling on untouched `2019` holdout	$\hat{p}_{\text{ens}}(\mathbf{x}) = \sum_{m=1}^{3} w_m \hat{p}_m(\mathbf{x})$, $w_m = \frac{\text{PR-AUC}_m}{\sum_j \text{PR-AUC}_j}$; $\text{Lift} = \frac{\text{AUPRC}}{\pi}$, $\text{BS} = \frac{1}{N}\sum(\hat{p}_k - y_k)^2$, ICI

Aim / RQ & Attribution Stage	Pre-Specified Design & Demarcation	Mathematical Estimator & Governance Gate
RQ2 · Steps 7–8: Consensus Gate $\mathcal{C}$ ★	Dual-model `TreeSHAP` (`Cat ∩ Best XGB`) + PySAT `Hitman(m22)` minimal hitting-set `AXP`	$\overline{\lvert\phi_j\rvert} \geq Q_{75} \cap \text{Supp}_{\text{AXP}}(f_j) \geq 0.05$ $\cap\ \text{Conf}(r) = \frac{N(r \wedge \hat{y}=1)}{N(r)} \geq 0.70$
RQ2 · Step 8: Intervention Rate (`IR`) ★	Step 8 tree/`AXP` counterfactual neutralization ($f_i \to x_{\text{med}}$); **never from FP-Growth**	$\text{IR}_{\Delta\hat{p}}(f_i) = \frac{1}{n}\sum_{k=1}^{n}\left(\hat{p}^{(k)} - \hat{p}^{(k)}_{f_i \to \text{med}}\right)$; $\text{IR}_{\text{AXP}}(f_i) = \frac{1}{\lvert E_1\rvert}\sum_{k \in E_1}\mathbb{I}(\hat{y}_{f_i \to \text{med}} = 0)$
RQ2 · Step 8: Interaction Effect (`IE`) ★	Risk-space contrast Δ_{11} & Class-1 minimal `AXP` co-occurrence lift (1,000 bootstrap CIs)	$\Delta_{11} = \mathbb{E}[\hat{p}_{11}] - \mathbb{E}[\hat{p}_{10}] - \mathbb{E}[\hat{p}_{01}] + \mathbb{E}[\hat{p}_{00}]$; $\text{IE}(f_i, f_j) = \frac{P(f_i \wedge f_j \lvert \text{AXP}_1)}{P(f_i \lvert \text{AXP}_1)P(f_j \lvert \text{AXP}_1)} > 1.0$
RQ3 · Step 9: Visual Analytics Demarcation	Training-cohort `bupaR`, `DTW`, & `FP-Growth`; **strictly excluded from predictor matrix**	$D_{\text{DTW}} = \min_{\pi \in \mathcal{A}_w}\sqrt{\sum \lVert \mathbf{x}_{i,t_1} - \mathbf{x}_{j,t_2}\rVert_2^2}$; $\text{Lift}_{\text{FPG}}(A \to B) = \frac{\text{Supp}(A \cup B)}{\text{Supp}(A)\text{Supp}(B)}$
RQ3 · Step 10: OODA Scenario Delta $\Delta\hat{p}$	Stateless counterfactual deprescribing ($\mathbf{x} \to \mathbf{x}'$); Imputation of Normality ($x_{\text{unobs}} = 0$)	$\Delta\hat{p} = \hat{p}_{\text{ens}}(\mathbf{x}') - \hat{p}_{\text{ens}}(\mathbf{x})$; $\text{Score}_{\text{comb}}(f) = 0.5 \cdot \text{SHAP}_{\text{norm}}(f) + 0.5 \cdot \text{FFA}_{\text{norm}}(f)$

Candidacy (`9 NOV`) vs. Final Defense (`23 NOV`) Scope Demarcation: Today's examination locks the pre-specified research questions (`RQ1-RQ3`), mathematical estimators, and strict predictive-vs.-visual boundary across Specific Aims 1–3. Complete empirical 2019 holdout outcomes, feature-level `IR` / `IE` rankings, and post-execution sensitivity findings are staged for the November 23 Final Dissertation Defense.

↩ Return to Consensus Attribution (Aim 2) | ↩ Return to Committee Discussion (Slide 10)